

MATTHEW COATES

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EDUCATION

- PhD**, Epidemiology, University of California, Los Angeles • Los Angeles, CA June 2026
Advisor: Onyebuchi A. Arah, MD DSc PhD MSc MPH
Dissertation: Augmenting graphical tools for causal inference in non-communicable and infectious disease epidemiology: applications in cancer and vaccine effectiveness studies
- MPH**, Global Health – Health Metrics and Evaluation, University of Washington • Seattle, WA 2016
Advisor: Haidong Wang, PhD MA
Thesis: Quantifying selection bias from birth history estimates of child mortality
- BA**, Neurobiology, minor in Global Health and Health Policy, Harvard University • Cambridge, MA 2013
Thesis: Association of socioeconomic status with prefrontal cortical thickness and subcortical volumes in children and adolescents

RESEARCH EXPERIENCE

- University of California, Los Angeles**, Los Angeles, California
Graduate Student Researcher 7-9/2023
- University of California, Los Angeles**, Los Angeles, California
Graduate Student Researcher 7-9/2022
- Brigham and Women's Hospital, Division of Global Health Equity**, Remote in Los Angeles, California
Research Specialist 9/2019 -
- Harvard Medical School, Department of Global Health and Social Medicine**
Program in Global Noncommunicable Disease and Social Change, Boston, Massachusetts
Associate 10/2016 - 8/2019
- Institute for Health Metrics and Evaluation**, Seattle, Washington
Post-Bachelor Fellow 8/2013-8/2016
- University Teaching Hospital, University of Zambia**, Lusaka, Zambia
MPH Practicum- Research Assistant SUMMER 2015

PUBLICATIONS

49. **Coates MM**, Rowe SL, Sullivan SG, Munoz-Rivas FM, Arah OA, & Regan AK. (2026). Assessing the use of medical insurance claims and electronic health records to measure COVID-19 vaccination during pregnancy. *Drug Safety*, 49: 567-580.
48. Ferrari G, Davies J, Thapa A, et al., including **Coates MM**. (2025). Prospective cohort study evaluating feasibility, acceptability, and clinical impact of diabetes self-management education in a PEN-Plus program in Southeastern Liberia. *PLOS Global Public Health*, 5(12): e0005657.

47. Rowe SL, Sullivan SG, Koerber JJ, Munoz FM, **Coates MM**, Arah OA, & Regan AK. (2025). Adverse events associated with COVID-19 vaccination or diagnosis among pregnant and non-pregnant women in the United States, 2021-2022. *International Journal of Infectious Diseases*, 160, 108050.
46. Rowe SL, Sullivan SG, Munoz FM, **Coates MM**, Agnew B, Arah OA, & Regan AK. (2025). COVID-19 vaccination during pregnancy and major structural birth defects. *Pediatrics*, 155(4), e2024069778.
45. Rowe SL, Sullivan SG, Munoz FM, **Coates MM**, Arah OA, & Regan AK. (2025). Uptake of recommended vaccines during pregnancy among publicly- and privately-insured people in the United States, December 2020 – September 2022. *American Journal of Public Health*, 115(3), 354-363.
44. Regan AK, Rowe SL, Sullivan SG, **Coates MM**, Munoz FM, & Arah OA. (2025). COVID-19 antiviral medication use among pregnant and recently pregnant US outpatients. *Clinical Infectious Diseases*, 80(3), 512-519.
43. Ferrari G, Boti M, Nakotwa D, Gomber A, **Coates MM**, et al. (2024). Clinical insights from a sub-study of a randomized control trial of continuous glucose monitor use in patients living with type 1 diabetes in rural Malawi. *Journal of Endocrinology, Metabolism, and Diabetes of South Africa*, 29(3), 87-93.
42. **Coates MM**, Arah OA, Matthews TA, Sandler DP, Jackson CL, and Li J. (2024). Multiple forms of perceived job discrimination and hypertension risk among employed women: Findings from the Sister Study. *American Journal of Industrial Medicine*, 67(9):844-856.
41. Gomber A, Valeta F, **Coates MM**, et al. (2024). Feasibility of continuous glucose monitoring in patients with type 1 diabetes at two district hospitals in Neno, Malawi: a randomised controlled trial. *BMJ Open*, 14:e075554.
40. Adler AJ, Wroe EB, Atzori A, et al., including **Coates MM**. (2024). Protocol for an evaluation of the initiation of an integrated longitudinal outpatient care model for severe chronic non-communicable diseases (PEN-Plus) at secondary care facilities (district hospitals) in 10 lower-income countries. *BMJ Open*, 14: e074182.
39. Ahmed S, Cao Y, Wang Z, **Coates MM**, et al. (2024). Service readiness for the management of non-communicable diseases in publicly financed facilities in Malawi: findings from the 2019 Harmonized Health Facility Assessment census survey. *BMJ Open*, 14: e072511.
38. Boudreaux C, Salvi D, Adler A, Wroe EB, **Coates MM**, Olsen M, Jain Y, Mocumbi AO, & Bukhman G. (2023). Realigning noncommunicable disease monitoring with health systems priorities in the Africa region. *Health Promotion International*, 38(6).
37. Adler AJ, Drown L, Boudreaux C, **Coates MM**, Marx A, Akala O, Waqanivalu T, Xu H, & Bukhman G. (2023). Understanding integrated service delivery: a scoping review of models for noncommunicable disease and mental health interventions in low-and-middle income countries. *BMC Health Services Research*, 23(99).
36. Matthews S, **Coates MM**, Bukhman A, et al. (2022). Health system capacity to manage diabetic ketoacidosis in nine low-income and lower-middle income countries: A cross-sectional analysis of nationally representative survey data. *eClinicalMedicine*, 55: 101759.
35. Haakenstad A, **Coates M**, Bukhman G, McConnell M, & Verguet S. (2022). Comparative health systems analysis of differences in the catastrophic health expenditure associated with non-communicable vs communicable diseases among adults in six countries. *Health Policy and Planning*, 37(9): 1107-1115.
34. **Coates MM**, Ezzati M, Robles Aguilar G, et al. (2021). Burden of disease among the world's poorest billion people: An expert-informed secondary analysis of the Global Burden of Disease estimates. *PLoS ONE*, 16(8): e0253073.
33. Gupta N, Mocumbi A, Arwal SH, et al., including **Coates MM** (as part of NCDI Poverty Network Secretariat). (2021). Prioritizing health-sector interventions for noncommunicable diseases and injuries in low- and lower-middle income countries: National NCDI Poverty Commissions. *Global Health Science and Practice*, 9(3).

32. **Coates MM**, Sliwa K, Watkins DA, et al. (2021). An investment case for the prevention and management of rheumatic heart disease in the African Union 2021-30: a modelling study. *Lancet Global Health*, 9(7): E957-E966.
31. Mwangi KJ, Gathecha G, Nyamongo M, et al. including **Coates M**. (2020) Reframing non-communicable diseases and injuries for equity in the era of universal health coverage: findings and recommendations from the Kenya NCDI Poverty Commission. *Annals of Global Health*, 87(1): 3, 1-16.
30. Boudreaux, C, Noble C, **Coates MM**, et al. (2020). Noncommunicable disease (NCD) strategic plans in low- and lower-middle income Sub-Saharan Africa: framing and policy response. *Global Health Action*, 13(1): 1805165.
29. **Coates MM**, Kintu A, Gupta N, et al. (2020). Burden of noncommunicable disease from infectious risks in 2017: a modelling study. *Lancet Global Health*, 8(12): E1489-E1498.
28. Gupta N, **Coates MM**, Bekele A, et al. (2020). Low availability of equipment and medications for non-communicable diseases and injuries at public first-referral level hospitals in eight low-income countries. *BMJ Open*, 10: e038842.
27. Bukhman G, Mocumbi AO, et al. including **Coates MM**. (2020). The *Lancet* NCDI Poverty Commission: bridging a gap in universal health coverage for the poorest billion. *The Lancet*, 396(10256), 991-1044.
26. Johansson KA, **Coates MM**, Økland J-M, Tsuchiya A, Norheim OF, Haaland Ø. (2020). Health by disease categories. In Cookson, R, Griffin, S, Norheim, O and Culyer, A. *Distributional Cost-Effectiveness Analysis: A Handbook of Equity-Informative Health Economic Evaluation*. Oxford University Press.
25. Johanssen KA, Økland J-M, Skaftun EK, Bukhman G, Norheim OF, **Coates MM**, Haaland Ø. (2020). Estimating health adjusted age at death (HAAD). *PLOS ONE* 15(7): e0235955.
24. **Coates MM**, Kamanda M, Kintu A et al. (2019). A comparison of all-cause and cause-specific mortality by household socioeconomic status across seven INDEPTH Network health and demographic surveillance systems in sub-Saharan Africa. *Global Health Action*, 12(1): 1608013.
23. Haakenstad A, **Coates M**, Marx A, Bukhman G, Verguet S. (2019). Disaggregating catastrophic health expenditure by disease area: cross-country estimates based on the World Health Surveys. *BMC Medicine*, 17: 36.
22. Liang J, Li X, Kang C, et al. including **Coates MM**. (2018). Maternal mortality ratios in 2852 Chinese counties, 1996-2015, and achievement of Millennium Development Goal 5 in China: a subnational analysis of the Global Burden of Disease Study 2016. *The Lancet*, 393(10168): 241-252.
21. Bui AL*, **Coates MM***, Matthay EC*. (2018). Years of life lost due to encounters with law enforcement in the USA, 2015-2016. *Journal of Epidemiology and Community Health*, 72(8): 715-718.
20. Rabiee R, Agardh E, **Coates MM**, Allebeck P, Danielsson A-K. (2017). Alcohol-attributed disease burden and alcohol policies in the BRICS-countries during the years 1990-2013. *J Global Health*, 7(1): 010404.
19. Gething PW, Casey DC, Weiss DJ, et al. including **Coates MM**. (2016). Mapping *Plasmodium falciparum* malaria mortality in Africa between 1990 and 2015. *New England Journal of Medicine*, 375(25): 2435-2445.
18. GBD 2015 Risk Factors Collaborators including **Coates MM**. (2016). Global, regional and national comparative risk assessment of 79 behavioural, environmental and occupational, and metabolic risks or clusters of risks in 195 countries, 1990-2015: a systematic analysis for the Global Burden of Disease Study 2015. *The Lancet*, 388(10053): 1659-1724.
17. GBD 2015 SDG Collaborators including **Coates MM**. (2016). Measuring the health-related Sustainable Development Goals in 188 countries: An analysis from the Global Burden of Disease Study 2015. *The Lancet*, 388(10053): 1813-1850.
16. GBD 2015 DALYs and HALE Collaborators including **Coates MM**. (2016) Global, regional, and national

disability-adjusted life years (DALYs) for 315 diseases and injuries and healthy life expectancy (HALE) for 195 countries and territories, 1990-2015: a systematic analysis for the Global Burden of Diseases, Injuries, and Risk Factors (GBD) 2015 Study. *The Lancet*, 388(10053): 1603-1658.

15. GBD 2015 Maternal Mortality Collaborators including **Coates MM**. (2016). Global, regional, and national levels of maternal mortality, 1990-2015: a systematic analysis for the Global Burden of Disease Study 2015. *The Lancet*, 388(10053): 1775-1812.

14. The Global Burden of Disease 2015 Child Mortality Collaborators including **Coates MM**. (2016). Global, regional, national and selected subnational levels of stillbirths, neonatal, infant, and under-5 mortality during 1980-2015: a systematic analysis for the Global Burden of Disease Study 2015. *The Lancet*, 388(10053): 1725-1774.

13. The Global Burden of Disease 2015 Mortality and Cause of Death Collaborators including **Coates MM**. (2016). Global, regional, and national life expectancy, all-cause and cause-specific mortality for 249 causes of death, 1980-2015: a systematic analysis for the Global Burden of Disease Study 2015. *The Lancet*, 388(10053): 1459-1544.

12. Mokdad AH, Forouzanfar MH, Daoud F, et al. including **Coates MM**. (2016). Health in Times of Uncertainty: a systematic analysis for the Global Burden of Disease Study 2013 in the Eastern Mediterranean Region, 1990-2013. *Lancet Global Health*, 4(10): e704-13.

11. Agardh E, Danielsson A-K, Ramstedt M, et al. including **Coates MM**. (2016). Alcohol-attributed disease burden in four Nordic countries: A comparison using the Global Burden of Disease, Injuries and Risk Factors 2013 study. *Addiction*, 111(10): 1806-13.

10. Kellerborg K, Danielsson A-K, Allebeck P, **Coates MM**, and Agardh E. (2016). Alcohol attributed disease burden: How methodological advances in the GBD 2013 study have changed the estimates in Sweden. *Scandinavian Journal of Public Health*, 44(6): 604-610.

9. The Global Burden of Disease 2015 HIV Collaborators including **Coates MM**. (2016). Estimates of global, regional, and national incidence, prevalence, and mortality of HIV, 1980-2015: the Global Burden of Disease Study 2015. *The Lancet HIV*, 3(8) e361-e387.

8. Wang Y, Li X, Zhou M, et al., including **Coates MM**. (2016). Under-5 mortality in 2851 Chinese counties, 1996-2012: a subnational assessment of achieving MDG 4 goals in China. *The Lancet*, 387(10015): 273-283.

7. Zhou M, Wang H, Zhu J, et al., including **Coates MM**. (2016). Cause-specific mortality for 240 causes in China during 1990-2013: a systematic subnational analysis for the Global Burden of Disease Study 2013. *The Lancet*, 387(10015): 251-272.

6. Newton JN, Briggs ADM, Murray CJL, et al., including **Coates MM**. (2015). Changes in health in England with analysis by region and areas of deprivation: findings of the Global Burden of Disease Study 2013. *The Lancet*, 386(10010): 2257-2274.

5. GBD 2013 Risk Factors Collaborators, including **Coates MM**. (2015) Global, regional and national comparative risk assessment of 79 behavioural, environmental and occupational, and metabolic risks or clusters of risks in 188 countries 1990-2013: a systematic analysis for the Global Burden of Disease Study 2013. *The Lancet*, 386(10010): 2287-2323.

4. GBD 2013 DALYs and HALE Collaborators, including **Coates MM**. (2015). Global, regional, and national disability-adjusted life years (DALYs) for 306 diseases and injuries and healthy life expectancy (HALE) for 188 countries, 1990-2013: quantifying the epidemiological transition. *The Lancet*, 386(10009): 2145-2191.

3. GBD 2013 Mortality and Causes of Death Collaborators, including **Coates MM**. (2014). Global, regional, and

national levels of age-sex specific all-cause and cause-specific mortality for 240 causes of death, 1990-2013: a systematic analysis for the Global Burden of Disease Study 2013. *The Lancet*, 385(9963):117-171.

2. Memish ZA, Jaber S, Mokdad AH, AlMazroa MA, Murray CJL, Al Rabeeah AA, and the Saudi Burden of Disease Collaborators, including **Coates MM**. (2014). Burden of Disease, Injuries, and Risk Factors in the Kingdom of Saudi Arabia, 1990–2010. *Preventing Chronic Disease*, 11, E169.

1. Wang H, Liddell CA, **Coates MM** et al. (2014). Global, regional, and national levels of neonatal, infant, and under-5 mortality during 1990-2013: a systematic analysis for the Global Burden of Disease Study 2013. *The Lancet*, 384(9947):957-79.

* Contributed equally

COMMENTARIES, EDITORIALS, AND LETTERS

2. Arah OA & **Coates MM**. (2025). Making DAGs even more useful: Using augmented causal diagrams to depict counterfactual, study design, measurement, analytical, and interventional features. *International Journal of Epidemiology*, 54(6): dyaf207.

1. **Coates MM**, Wolock CJ, Arah OA. (2026). RE: Prediagnostic exposures and cancer survival: Can a meaningful causal estimand be specified? *Epidemiology*, 37(1):p e1-e2.

MANUSCRIPTS UNDER REVIEW

1. **Coates MM**, Arah OA, Rowe SL, Sullivan SG, Muñoz FM, & Regan AK. Risk of severe COVID-19 in pregnant and postpartum women after the pandemic.

2. Rowe SL, **Coates MM**, Sullivan SG, Koerber JJ, Muñoz FM, Arah OA, & Regan AK. Does vaccine timing matter? Effectiveness of COVID-19 vaccine during pregnancy against medically-attended maternal COVID-19.

MANUSCRIPTS IN PREPARATION

1. **Coates MM** et al. An investment case for expanding health services for severe, chronic non-communicable diseases in the WHO African Region between 2025-2035 through the PEN-Plus Strategy: a modeling study.

2. **Coates MM**, Hazlett C, Zhang ZF, Nianogo RA, Regan AK, Wolock CJ, & Arah OA. Understanding lead-time bias in cancer screening studies through augmented causal diagrams and causal estimands.

3. Coons W*, **Coates MM***, & Arah OA. Transporting average causal effects across observational settings.

4. GBD 2023 fire, heat, and hot substances collaborators, including **Coates MM**. Global burden of injuries from fire, heat, and hot substances, 1990-2023: a systematic analysis for the Global Burden of Disease Study 2023.

5. **Coates MM**, Nianogo RA, & Arah OA. Towards an algorithm for translating causal loop diagrams to directed acyclic graphs with applications to cancer and infectious disease research.

AWARDS, HONORS, AND FELLOWSHIPS

University of California, Los Angeles

Carolbeth Korn Scholar Award, FSPH Most Outstanding Graduating Student	2026
UCLA Jonsson Comprehensive Cancer Center Trainee Conference Travel Award	2025
National Cancer Institute Ruth L. Kirschstein NRSA Institutional Research Training Grant (T32)	
UCLA Cancer Epidemiology Training Program Trainee	2023-2026
Ursula Mandel Scholarship	2023-2024
Celia and Joseph Blann Fellowship	2023-2024
Graduate Research Mentorship	2023-2024
Department Fellowship in Epidemiology	2021-2023
Graduate Dean's Scholar Award	2021-2023

University of Washington

Distinguished Master's Thesis Award for Biology and Life Sciences	2017
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Harvard University

High Honors, Undergraduate Thesis	2013
Cum Laude	2013
Robert C. Byrd Honors Scholarship, US Office of Postsecondary Education and Georgia Department of Education	2009

Conference Awards

First Place Poster, Behavioral Research Advancements in Neuroscience (BRAIN) Summer Symposium, Center for Behavioral Neuroscience, Atlanta, GA,	2012
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CONFERENCE PRESENTATIONS AND POSTERS

*Indicates presenter

14. Regan AK*, **Coates M**, Rowe SL, Sullivan SG, Munoz FM, & Arah OA. Post-pandemic cohort study to reassess the severity of SARS-CoV-2 infection during pregnancy. **Poster presented** at European Perinatal and Paediatric Epidemiology Conference. Oslo, Norway, September, 2025.
13. Regan AK*, **Coates MM**, Rowe SL, Sullivan SG, Munoz FM, & Arah OA. Incidence & severity of post-pandemic SARS-CoV-2 infection during pregnancy. **Poster presented** at IDWeek 2025 (joint annual meeting of the Infectious Diseases Society of America, the Society for Healthcare Epidemiology of America, the HIV Medicine Association, the Pediatric Diseases Society, and the Society of Infectious Diseases Pharmacists). Atlanta, GA, October 2025.
12. **Coates MM*** & Arah OA. Framing and extending immortal time bias with augmented causal diagrams and annotated causal estimands: immune time versus immune study population. **Oral presentation** at the Society for Epidemiologic Research Annual Meeting. Boston, MA, June 2025.
11. **Coates MM***, Anukam D, Zhang ZF, & Arah OA. Collider bias in estimating the effects of pre-incident-cancer exposures on post-incident-cancer outcomes. **Poster presented** at the Society for Epidemiologic

Research Annual Meeting. Boston, MA, June 2025.

10. **Coates MM***, Anukam D, Zhang ZF, & Arah OA. Identifying the effects of pre-diagnosis exposures on post-diagnosis outcomes among people with cancer. **Poster presented** at the University of California, Los Angeles Jonsson Comprehensive Cancer Center Annual Symposium. Los Angeles, CA, May 2025.
9. **Coates MM***, Zhang ZF, & Arah OA. Collider bias in estimating the effect of cancer screening on post-incident outcomes. **Poster presented** at the American Society of Preventive Oncology Annual Meeting. Philadelphia, PA, April 2025.
8. **Coates MM***, Rowe SL, Sullivan SG, Munoz FM, Arah OA, & Regan AK. Assessing the use of medical insurance claims and electronic health records to measure COVID-19 vaccination during pregnancy. **Poster presented** at the Society for Epidemiologic Research Mid-Year Meeting. Online, February 2025.
7. Regan AK*, Rowe SL, Sullivan SG, Munoz F, Griffith J, **Coates M**, & Arah OA. Modification of the relationship between prenatal COVID-19 vaccination and ankyloglossia by insurance type. **Poster presented** at the Society for Pediatric and Perinatal Epidemiologic Research Annual Meeting. Austin, TX, June 2024.
6. Rowe SL*, Sullivan S, Munoz FM, Griffith J, **Coates M**, Arah OA, & Regan AK. Neonatal outcomes among people vaccinated against COVID-19 during pregnancy in the United States. **Poster presented** at the Society for Epidemiologic Research Annual Meeting. Austin, TX, June 2024.
5. Rowe SL*, Sullivan S, Munoz FM, Arah O, **Coates M**, & Regan AK. Disparities in the uptake of recommended vaccines among publicly- and privately-insured pregnant people in the United States. **Poster presented** at the American Public Health Association Annual Meeting. Atlanta, GA, November 2023.
4. **Coates MM***, Arah OA, Matthews TA, Sandler DP, Jackson CL, & Li J. (2023). Multiple forms of job discrimination and hypertension risk among employed women: Findings from the Sister Study. **Oral presentation** at the Society for Epidemiologic Research Annual Meeting 2023, Portland, OR, June 2023.
3. Bui AL*, **Coates MM**, & Matthay EC. (2017). Quantifying years of life lost due to encounters with law enforcement in the United States, 2015-2016. **Poster presented** at California Medical Association Conference 2017, Anaheim, CA, October 2017.
2. Bui AL*, **Coates MM**, & Matthay EC. (2017). Quantifying years of life lost due to encounters with law enforcement in the United States, 2015-2016. **Poster presented** at American College of Physicians Southern California 2017 Meeting. Marina del Rey, CA, September 2017.
1. Bommer CE*, **Coates MM***, Diaz MM*, Pallis TT*. (2011). Behavioral and neural plasticity following appendage loss in crayfish (*Procambarus clarkii*). **Poster presented** at Behavioral Research Advancements in Neuroscience (BRAIN) 2011 Research Symposium. Center for Behavioral Neuroscience. Atlanta, GA, July 2011.

TEACHING AND MENTORSHIP

- *Teaching Assistant*, Epidemiology M204: Logic, Causation, and Probability, UCLA, Fall 2025 (25% FTE)
- *Teaching Assistant*, Epidemiology M211: Statistical Methods for Epidemiology, UCLA, Winter 2024 (25% FTE)
- *Teaching Assistant*, Epidemiology M204: Logic, Causation, and Probability, UCLA, Fall 2024 (25% FTE)
- *Teaching Assistant*, Epidemiology M204: Logic, Causation, and Probability, UCLA, Fall 2023 (25% FTE)
- *Mentorship Committee Co-Chair*, Epidemiology Student Association, UCLA, 2023-2024 academic year
- *Mentor*, Epidemiology Student Association Mentorship Program, UCLA, 2022-2024

- *Lecturer*, Cardiovascular disease and global health equity, Continuing Medical Education course. Harvard Medical School, February 29, 2020.
- *Workshop instructor*, NCDI Poverty Commission Initiators Workshop. Dubai, December 5-8, 2018.
- *Lecturer*, Cardiovascular disease and global health equity, Continuing Medical Education course. Harvard Medical School, July 28, 2018.

PROFESSIONAL MEMBERSHIPS

- California Center for Population Research, Student Affiliate, UCLA, 2022-present
- Society for Epidemiologic Research, member, 2022-present

PROFESSIONAL SERVICE

Manuscript Peer Review

- Nature Communications, 2025
- Advances in Public Health, 2025
- International Journal of Epidemiology, 2024-2025
- AIMS Public Health, 2024
- Global Health Action, 2020-2023
- The Lancet Global Health, 2022

ADDITIONAL TRAINING

- Modelers and Storytellers: Transdisciplinary Training to Advance Community Health Intervention Research, 3-week training program on systems science, data science, and community health science, UCLA, July 2023

COMPUTER SKILLS

- **Programming (order of experience):** R, Stata, SAS, SQL, Python, VBA in Excel
- **Code Management and Version Control:** Git (GitHub, SourceTree/Bitbucket)
- **Microsoft Office:** PowerPoint, Word, Excel, Outlook, Access